

Notes

Posterior mean genetic liability from family history

Model, sequential approximation, and evidence in `ltpred`

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<https://github.com/bvilhjal/ltpred> · v0.3.0

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Abstract

The LT-FH family of methods replaces a case/control label with $\mathbb{E}[a_i \mid \text{family data}]$ under a Gaussian liability-threshold model. That is ordinary BLUP after the liabilities have been replaced by their truncated-normal means; the published names (LT-FH, LT-FH++, ADuLT, PA-FGRS) are different choices of the truncation region C_F , not different genetic models. These notes fix the estimand, the observation-model menu, the exactness boundary of the Pearson–Aitken (PA) approximation, and the sampling conditions under which a pedigree Haseman–Elston fit recovers h^2 . Simulation under the model’s own generative assumptions gives $\text{Corr}(\text{PA}, \text{Gibbs}) \geq 0.9977$ without the censoring mixture, a classic-LT-FH causal-SNP noncentrality ratio of $1.47 \pm 0.04\times$ over case/control (independent SNPs), and a paired $+0.145 \pm 0.015$ LT-FH++ increment over matched ADuLT when onset is a deterministic function of liability and the CIP is known. Those last two figures are upper bounds relative to register data. They are not estimates of clinical utility.

The intended reader already uses GWAS, liability-scale h^2 , and Falconer’s threshold construction. The notes do not re-derive those. They are about what is being conditioned on, where the sequential approximation is exact, and which simulation claims survive their generative assumptions.

1 Estimand

Write a_i for the additive genetic liability of a designated proband, $\text{Var}(a_i) = h^2$ on the usual unit-liability scale, and ℓ_F for the vector of full liabilities on the people whose disease records are used (the proband may or may not be among them). The data determine a rectangle C_F —one interval per person—and the target is

$$\mu_i = \mathbb{E}[a_i \mid \ell_F \in C_F, A, h^2, K(\cdot)], \quad (1)$$

where A is the additive relationship matrix and $K(\cdot)$ is the prevalence or cumulative-incidence model that turns status and age into interval endpoints.

If ℓ_F were observed continuously, (1) would be the selection index / animal-model BLUP

$$\mathbb{E}[a_i \mid \ell_F] = \text{Cov}(a_i, \ell_F) \text{Var}(\ell_F)^{-1} \ell_F \quad (2)$$

[4, 5]. A single relative with relationship r contributes $r h^2 \ell_r$. Disease records do not give ℓ_F . They give $\ell_F \in C_F$. Under joint normality,

$$\mathbb{E}[a_i \mid \ell_F \in C_F] = \text{Cov}(a_i, \ell_F) \text{Var}(\ell_F)^{-1} \mathbb{E}[\ell_F \mid \ell_F \in C_F]. \quad (3)$$

The BLUP weights are unchanged; only the right-hand side is replaced by a truncated-normal mean. The map from the binary (or censored) records to μ_i is therefore nonlinear, and reduces to (2) only when every interval collapses to a point.

Two immediate consequences, both easy to get wrong in an analysis.

1. **The proband’s own diagnosis is optional.** Include it when μ_i is a GWAS phenotype *constructed from* that diagnosis [7, 18]. Leave the corresponding interval as $(-\infty, \infty)$ when the same diagnosis is the outcome you will later predict or classify. Conditioning on the answer is leakage, not sophistication.
2. **(1) is not a SNP polygenic score.** No marker effects enter. Combining μ_i with a PGS is a downstream model [8, 13].

The multivariate threshold model on a pedigree is Sham’s framework [21]. So, Kwan, Cherny and Sham [22] already combined family history with known loci, age, and competing mortality; they targeted absolute risk. Here the target is the breeding value, used as a phenotype.

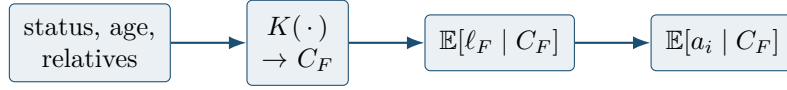


Figure 1: The published methods differ in how they build C_F . They share (3).

2 Covariance

The liability split is the usual one,

$$\ell = a + e, \quad a \sim \mathcal{N}(0, h^2), \quad e \sim \mathcal{N}(0, 1 - h^2), \quad a \perp\!\!\!\perp e, \quad (4)$$

so $\ell \sim \mathcal{N}(0, 1)$ and a case is $\ell > T$ with $T = \Phi^{-1}(1 - K)$ at a single prevalence K [23, 3]. On a pedigree,

$$\text{Cov}(a_i, a_j) = h^2 A_{ij}, \quad \text{Cov}(\ell_i, \ell_j) = h^2 A_{ij} \ (i \neq j), \quad \text{Var}(\ell_i) = 1, \quad (5)$$

with $A_{ij} = 2\phi_{ij}$. Mates are taken as unrelated ($A_{mf} = 0$): the default A is not a generative model of assortment. Inbreeding is handled by the tabular method [6], after which full liabilities are rescaled to unit variance so that $\Phi^{-1}(1 - K)$ still means prevalence K .

Shared environment is optional and off-diagonal,

$$\text{Cov}(\ell_i, \ell_j) = h^2 A_{ij} + c^2 C_{ij} + m^2 M_{ij}, \quad h^2 + c^2 + m^2 \leq 1. \quad (6)$$

The target a_i still couples to relatives only through $h^2 A$. Identification is the usual contrast arithmetic: C (full-sib environment) is the excess of sib covariance over parent–offspring (both have $A = \frac{1}{2}$); M (couple) is the covariance of genetically unrelated mates, and therefore confounds household sharing with assortment [20]. A directional maternal effect is not a symmetric kernel and is not M . An environment that scales with A is absorbed into h^2 and is not estimable. Nuclear families give roughly three contrasts; a bank of relationship-specific environments needs the relative types that extended pedigrees supply. This is the pedigree analogue of why ACE is identified from MZ/DZ twins and $A + C + D$ is not.

One inherited LTFHPlus convention is worth knowing before comparing constructors: two same-side half-sibs in the role grammar are related $\frac{1}{2}h^2$ to *each other* (implied shared second parent). A pedigree in which they have distinct other parents must be built from A , not from those role labels.

3 The observation-model menu

LT-FH, LT-FH++ and ADuLT are names for C_F , not for an inference algorithm. Gibbs and PA both apply to the first three. PA-FGRS, in the published specification and in the implementation, is PA-specific.

Classic LT-FH [7]. One K , one T . Cases $[T, \infty)$, controls $(-\infty, T]$, relatives included. This is the configuration-specific successor of GWAX [15].

LT-FH++ and ADuLT [18, 19]. Age, sex and cohort enter only through a CIP $K(t; s, b)$ and

$$T_i = \Phi^{-1}(1 - K(t_i; s_i, b_i)). \quad (7)$$

A control at current age c_i is truncated above at $T_i(c_i)$. A case with onset a_i is *pinned* at $T_i(a_i)$ (lower = upper). The same T_i with relatives is LT-FH++; without them, ADuLT.

Pinning is the strong assumption in this construction, and it is easy to understate. It says that the CIP is an error-free map from liability to onset, that onset dates are exact, and that two cases with the same stratum and onset have identical liability—zero residual variance given a_i . The inverse map is then the identity: the observation *is* the liability. When onset is a window, or K is estimated, an interval $[T_i(a_i), \infty)$ or the lifetime interval $[T_{\text{pop}}, \infty)$ is the more honest likelihood.

Because the BLUP weights in (3) do not depend on the T_i , a shared threshold error is mostly a calibration shift. A stratum-specific error also reorders scores.

Base PA-FGRS [12]. Observed cases use the *lifetime* interval $[\Phi^{-1}(1 - K_{\text{pop}}), \infty)$. Age enters for censored controls through the mixture in Section 4.3, not through the case threshold. Helpers that put cases above $T_i(a_i)$ are a different observation model; they are not PA-FGRS_{ADT}. Kendler’s FGRS [10] is a third thing and is not implemented.

PGS and μ_i . Split $a = s + u$ with $\text{Var}(s) = p := h_{\text{SNP}}^2/h^2$. If two standardised scores satisfy $\text{PGS} \perp\!\!\!\perp \mu \mid a$, then

$$\text{Corr}(\text{PGS}, \mu) = a_{\text{PGS}} b_{\mu} \sqrt{p} = \sqrt{R_{\text{PGS}}^2 R_{\mu}^2 p}. \quad (8)$$

The scores can each be genuinely predictive of a and still almost uncorrelated, because each is mostly noise relative to a [8, 13]. Non-overlapping discovery and family samples remove one shared noise term. They do not imply the residual is zero (ancestry, assortment, indirect effects, selection).

4 Computing the truncated mean

There is no closed form for $\mathbb{E}[\ell_F \mid C_F]$ once several intervals are live. Two approximations are used.

4.1 Gibbs

Coordinate-wise inverse-CDF sampling of the conditional normals [11], with coefficients from the precision $Q = \Sigma^{-1}$: $\text{sd}_j = \sqrt{1/Q_{jj}}$, $P_{ij} = -Q_{ij}/Q_{jj}$. Pinned coordinates are held. Far-tail intervals cannot be drawn on the probability scale (Φ underflows near $|z| \approx 38.5$); those use a log-scale Rayleigh construction. Σ must be strictly positive definite.

The Monte Carlo standard error is the batch-means estimator [9] that LTFHPlus takes from **batchmeans**. Families that share a structure share one factorisation of Σ ; the rest is embarrassingly parallel. This is the sampling-based reference for every no-mixture claim below.

4.2 Pearson–Aitken

If one coordinate of a multivariate normal is selected from $\mathcal{N}(m_i, v_i)$ to (m^*, v^*) , the others update by [17, 1]

$$m_j \leftarrow m_j + \frac{\Sigma_{ji}}{v_i}(m^* - m_i), \quad \Sigma_{jk} \leftarrow \Sigma_{jk} + \frac{\Sigma_{ji}\Sigma_{ik}}{v_i^2}(v^* - v_i). \quad (9)$$

Mendell and Elston applied this sequentially to threshold traits [16]. The implementation folds observed members last-to-first, then applies the target’s own interval to the updated $\mathcal{N}(m_0, v_0)$. For the genetic target the last step is a no-op (bounds $(-\infty, \infty)$). For $\mathbb{E}[\ell_o \mid C_F]$ it is the same estimand Gibbs uses.

After one truncation the selected law is no longer Gaussian. PA keeps the first two moments and proceeds as if it were. Hence: exact for a single interval or a pin; a sequential two-moment approximation thereafter. Fold order changes the error. Canonicalising to a sorted role set makes the result a function of the pedigree shape, not of input row order—reproducibility, not accuracy.

On the no-mixture structures we have tested, that approximation error is small enough to ignore for the posterior *mean* (Section 6). It has not been characterised for the mixture, and PA’s reported conditional variance is itself an approximation. A PA standard error of zero means no Monte Carlo error.

4.3 Censored-control mixture

A person unaffected only up to current follow-up is a mixture of a lifetime control and a not-yet-onset future case [12]. With current CIP K_i and lifetime K_{pop} ,

$$\pi = \frac{\Phi_{\text{below}}}{\Phi_{\text{below}} + (1 - \Phi_{\text{below}})(K_{\text{pop}} - K_i)/K_{\text{pop}}}, \quad (10)$$

where the split is the *lifetime* threshold and Φ_{below} is the current conditional mass below it. The selected moments are the corresponding two-component mean and variance. Age enters only through K_i .

The factor $(K_{\text{pop}} - K_i)/K_{\text{pop}}$ is the CIP probability that a future case has not yet onset. That is an onset-timing assumption: among people who will eventually be cases, time-to-onset is independent of liability, and censoring is non-informative given the stratum. If higher liability advances onset, the not-yet-onset component is not the above-threshold tail assigned to it. There is no Gibbs implementation of (10), so this part of the model has no sampling-based check in the package.

5 Fitting h^2 is a different problem

Prediction *conditions* on h^2 . Bring a liability-scale estimate; if what you have is observed-scale (OLS / GREML / LDSC on 0/1), Lee et al. [14] give

$$h_{\text{liab}}^2 = h_{\text{obs}}^2 \cdot \frac{K(1 - K)}{z^2} \cdot \frac{K(1 - K)}{P(1 - P)}, \quad z = \phi(\Phi^{-1}(1 - K)), \quad (11)$$

with P the sample case fraction ($P = K$ drops the second factor).

A pedigree fit of h^2 from the same family records is available, but it is not “LT-FH++ with h^2 free.” The implementation is data augmentation: draw $\ell \mid C_F, \Sigma(h^2)$, then a damped Haseman–Elston step

$$\widehat{h^2} = \frac{\sum_{\text{pairs}} A_{ij} \ell_i \ell_j}{\sum_{\text{pairs}} A_{ij}^2}. \quad (12)$$

This is a stochastic-approximation fixed point, not a posterior. The number it reports as a standard error is within-chain Monte Carlo noise. Across-cohort SD is 20–30× larger (Table 3).

Law of total expectation recovers $\mathbb{E}[\ell_i \ell_j] = h^2 A_{ij}$ from the augmented products only for independent, non-overlapping, unascertained families and a *common* T per trait. Personalised or onset-pinned rectangles—even the internally coherent ones that prediction wants—drive (12) to the $h^2 \sim 1$ boundary and invent a C component. Those inputs are refused. Ascertainment is not cured by more sweeps; it is a missing sampling likelihood. The same remarks apply to the multiple HE fit of (h^2, c^2, m^2) . A family bootstrap, when clusters are independent, estimates sampling variability under the fitted design. It does not de-bias selection.

If the question is h^2 from an ascertained case/control GWAS, use LDSC or GREML and (11). If the question is h^2 from population-sampled nuclear families with one threshold, the pedigree HE is a reasonable moment estimator and is calibrated in that design (Section 6). Those are not the same analysis.

6 Evidence

Unless noted, $\pm$ is the SE of a mean across independent simulated cohorts. The numbers below are from committed benchmark artifacts (31 July and 1 August 2026). They do not automatically validate later code. Every GWAS panel uses independent SNPs; a real-LD run has not been done. There is no locked comparison to R LTFHPlus.

Two ratios appear and are not interchangeable. An *NCP ratio* is $(\overline{\chi^2}_{\text{causal}} - 1)$ over the same quantity for case/control. An *eff- N proxy* is a squared-correlation ratio to the case/control label.

6.1 The sequential approximation

On a 27-cell grid of classic LT-FH rectangles (three family structures $\times$ three $h^2 \times$ three K ; five cohorts of 1,500 families), $\text{Corr}(\text{PA}, \text{Gibbs})$ is 0.9977–0.9999. Stress pedigrees remain ≥ 0.9984 . Table 1 is the $h^2 = 0.5$, $K = 0.05$ slice. Calibration slopes of correctly specified PA sit on 1 to within simulation error (e.g. 1.006 ± 0.008 at $K = 0.05$, parents + two sibs). That is the argument for using PA as the default engine when the mixture is off. It is not a licence to treat PA and Gibbs as interchangeable once (10) is on.

At four threads on one machine the object-path PA sweep is 203–492 $\times$ faster than grouped Gibbs (2,000 families, 25,000 draws). The ratio moves with thread count because Gibbs is the parallel engine. The useful statement is that the approximation is cheap enough for biobank-scale C_F , not a hardware-independent constant.

Table 1: PA versus Gibbs, classic LT-FH bounds, $h^2 = 0.5$, $K = 0.05$, five seeds. Correlation is with simulated a_i . The eff- N proxy is not an NCP ratio.

Structure	corr. Gibbs	corr. PA	eff- N / c-c	Corr(PA, Gibbs)
parents	0.382 ± 0.009	0.382 ± 0.009	$1.36 \pm 0.03\times$	0.9997
parents + 2 sibs	0.437 ± 0.009	0.437 ± 0.009	$1.66 \pm 0.07\times$	0.9997
extended	0.419 ± 0.014	0.420 ± 0.014	$1.71 \pm 0.09\times$	0.9997

6.2 Classic LT-FH as a GWAS phenotype

Three genotype/effect/cohort replicates; 10,000 probands, 5,000 independent SNPs, 30 causal, $h^2 = 0.5$, $K = 0.05$, parents plus one sibling (Table 2). The same classic LT-FH C_F , inferred by either engine, gives an NCP ratio of $1.47 \pm 0.04\times$ over case/control, with λ_{GC} consistent with 1. A single-seed 1.53 $\times$ does not survive replication. The oracle (true a_i) is 8.8 $\times$: most of the liability is still unused, as it should be when $h^2 = 0.5$ and relatives are few.

Table 2: Independent-SNP GWAS. NCP uses $\overline{\chi^2} - 1$. Power is the fraction of causal SNPs at $p < 5 \times 10^{-8}$.

Phenotype	mean causal χ^2	NCP / c-c	power	λ_{GC}
case/control	39.63 ± 2.19	$1.00\times$	$41.1 \pm 2.2\%$	1.011 ± 0.009
LT-FH (Gibbs)	57.53 ± 1.87	$1.468 \pm 0.040\times$	$48.9 \pm 4.0\%$	1.012 ± 0.019
LT-FH (PA)	57.60 ± 1.91	$1.469 \pm 0.039\times$	$48.9 \pm 4.0\%$	1.006 ± 0.021
oracle a_i	337.37 ± 2.07	$8.77 \pm 0.57\times$	$75.6 \pm 1.1\%$	1.047 ± 0.015

6.3 What personalisation is worth

The integrated LT-FH++ panel uses age/sex/cohort CIP, coherent follow-up, competing death, ascertainment, and stratified null SNPs. Onset is the CIP inverse of true ℓ , and cases are pinned at that T_i —so a case’s full liability is essentially observed. Ten paired replicates.

Matched ADuLT (same personalised T_i on the proband, no relatives) reaches an adjusted NCP ratio of $1.049 \pm 0.004\times$. Full LT-FH++ reaches $1.194 \pm 0.006\times$. The paired increment is $+0.145 \pm 0.015$ (95% CI half-width). Calibration slope after covariate adjustment is 0.995 ± 0.015 . A sex-only panel closes a female–male mean-score-error gap of 0.051 ± 0.001 ; the adjusted power increment is consistent with zero (0.004 ± 0.005).

Read this as: under a known CIP and a deterministic onset map, family history still adds association signal after the proband’s own T_i is personalised, and most of that increment is already present once age is in the CIP. Do not read it as the increment you will see in a register, where onset is noisy and $K(t; s, b)$ is estimated. The pin-equals-liability identity is an assumption of the observation model, and the simulation honours it.

6.4 Complementarity with a PGS

Five replicates, 10,000 probands, 2,000 independent SNPs, 50/50 train/test, PGS trained on the train case/control GWAS (infinitesimal weights; no LD to shrink). Against held-out a_i , test R^2 is 0.236 ± 0.009 (PGS), 0.170 ± 0.003 (classic LT-FH), 0.339 ± 0.009 (OLS on both). Incremental R^2 : $+0.104 \pm 0.007$ for LT-FH over PGS, $+0.169 \pm 0.019$ for PGS over LT-FH. Observed $\text{Corr}(\text{PGS}, \mu) = 0.2009 \pm 0.0046$ against (8) at $p = 1$ (0.2003 ± 0.0050). Conditional independence holds by construction here. That is a verification of the identity, not a test of the failure modes.

6.5 Moment recovery of h^2 and C

Twenty-five unascertained population cohorts, 3,000 families, parents + two sibs, $K = 0.10$, one threshold (Table 3). Bias is small relative to across-cohort SD. The fitter’s Monte Carlo SE is not a sampling SE.

Table 3: Haseman–Elston data-augmentation fit of liability-scale h^2 under the design it is identified in.

true h^2	mean	bias	emp. SD	MC SE	SD / MC SE
0.2	0.201	+0.001	0.037	0.0020	19×
0.4	0.387	−0.013	0.055	0.0022	25×
0.6	0.577	−0.023	0.063	0.0023	28×
0.8	0.786	−0.014	0.061	0.0026	24×

Joint $A + C$ on the same design (mean, emp. SD): true (0.4, 0.2) recovers 0.399 (0.057), 0.190 (0.034); true (0.6, 0) recovers 0.583 (0.057) and a boundary 0.013 (0.012). A point near

zero is not a Type I rate. Omitting a real C is mostly a problem for interpreting h^2 (at true $c^2 = 0.3$, additive-only $\widehat{h^2} \approx 0.75$ versus ≈ 0.48 with C in the model), not for predicting a_i .

A research-only cross-trait HE, 25 cohorts, recovers $r_g = 0$ as -0.006 (SD 0.080) in the presence of phenotypic correlation 0.2, and $r_g = 0.6$ as 0.598 (SD 0.071).

7 What this does not do

The score is the additive-model projection of the family history you fed it. Household transmission, dominance, epistasis, indirect genetic effects, and generative assortment are not in A . When they are in the data, they are in μ_i .

Diagnosis and family-history reporting are part of the observation model. Parental-history Alzheimer GWAS is a documented example of survival and participation bias [24]; shallow psychiatric phenotypes can carry heritable confounding [2]. A CIP estimated from an ascertained biobank sample is the wrong $K(\cdot)$. Kaplan–Meier estimates net risk under independent censoring; competing death needs Aalen–Johansen. The mixture (10) inherits both requirements.

Related target families produce related μ_i by construction. A linear GWAS of those scores without a mixed model, or without pruning to non-overlapping families, is the analysis Zhuang et al. [25] warned about.

Three evidence gaps would change how hard Section 6 should be leaned on: a locked vector comparison to LTFHPlus, a real-LD GWAS, and a Gibbs implementation of (10).

Software. `ltpred` is a Python implementation of the construction above (NumPy/SciPy; optional Numba). It does not run a GWAS. Heavier prototypes—sex-limited A , genetic nurture, onset-age decay of r_g , a common-factor summary of r_g , a Monte Carlo EM likelihood—are checkout-only and are not covered here. Source: <https://github.com/bvilhjal/ltpred>. Cite the method actually used [7, 18, 19, 12] and the software.

References

- [1] A. C. Aitken. Note on selection from a multivariate normal population. *Proceedings of the Edinburgh Mathematical Society*, 4(2):106–110, 1935. doi:10.1017/S0013091500008063.
- [2] N. Cai et al. A Perspective on heritable confounding in psychiatric genetics. *Nature Genetics*, 2026. doi:10.1038/s41588-025-02465-y.
- [3] D. S. Falconer. The inheritance of liability to certain diseases, estimated from the incidence among relatives. *Annals of Human Genetics*, 29(1):51–76, 1965.
- [4] L. N. Hazel. The genetic basis for constructing selection indexes. *Genetics*, 28(6):476–490, 1943.
- [5] C. R. Henderson. Best linear unbiased estimation and prediction under a selection model. *Biometrics*, 31(2):423–447, 1975.
- [6] C. R. Henderson. A simple method for computing the inverse of a numerator relationship matrix used in prediction of breeding values. *Biometrics*, 32(1):69–83, 1976.
- [7] M. L. A. Hujoel, S. Gazal, P.-R. Loh, N. Patterson, and A. L. Price. Liability threshold modeling of case–control status and family history of disease increases association power. *Nature Genetics*, 52(5):541–547, 2020. doi:10.1038/s41588-020-0613-6.
- [8] M. L. A. Hujoel, P.-R. Loh, B. M. Neale, and A. L. Price. Incorporating family history of disease improves polygenic risk scores in diverse populations. *Cell Genomics*, 2(7):100152, 2022. doi:10.1016/j.xgen.2022.100152.

- [9] G. L. Jones, M. Haran, B. S. Caffo, and R. Neath. Fixed-width output analysis for Markov chain Monte Carlo. *Journal of the American Statistical Association*, 101(476):1537–1547, 2006.
- [10] K. S. Kendler, H. Ohlsson, J. Sundquist, and K. Sundquist. The family genetic risk score. *JAMA Psychiatry*, 78(6):637–643, 2021. doi:10.1001/jamapsychiatry.2021.0336.
- [11] J. H. Kotecha and P. M. Djurić. Gibbs sampling approach for generation of truncated multivariate Gaussian random variables. In *ICASSP*, 1999.
- [12] M. Dybdahl Krebs et al. Genetic liability estimated from large-scale family data improves genetic prediction, risk score profiling, and gene mapping for major depression. *The American Journal of Human Genetics*, 111(11):2494–2509, 2024. doi:10.1016/j.ajhg.2024.09.009.
- [13] M. Dybdahl Krebs et al. The relationship between genotype- and phenotype-based estimates of genetic liability to psychiatric disorders, in practice and in theory. *The American Journal of Human Genetics*, 113(1):184–201, 2026. doi:10.1016/j.ajhg.2025.11.016.
- [14] S. H. Lee, N. R. Wray, M. E. Goddard, and P. M. Visscher. Estimating missing heritability for disease from genome-wide association studies. *The American Journal of Human Genetics*, 88(3):294–305, 2011.
- [15] J. Z. Liu, Y. Erlich, and J. K. Pickrell. Case-control association mapping by proxy using family history of disease. *Nature Genetics*, 49(3):325–331, 2017. doi:10.1038/ng.3766.
- [16] N. R. Mendell and R. C. Elston. Multifactorial qualitative traits: genetic analysis and prediction of recurrence risks. *Biometrics*, 30(1):41–57, 1974.
- [17] K. Pearson. Mathematical contributions to the theory of evolution. XI. On the influence of natural selection on the variability and correlation of organs. *Philosophical Transactions of the Royal Society A*, 200:1–66, 1903.
- [18] E. M. Pedersen et al. Accounting for age of onset and family history improves power in genome-wide association studies. *The American Journal of Human Genetics*, 109(3):417–432, 2022. doi:10.1016/j.ajhg.2022.01.009.
- [19] E. M. Pedersen et al. ADuLT: an efficient and robust time-to-event GWAS. *Nature Communications*, 14:5553, 2023. doi:10.1038/s41467-023-41210-z.
- [20] M. R. Robinson et al. Genetic evidence of assortative mating in humans. *Nature Human Behaviour*, 1:0016, 2017.
- [21] P. Sham. *Statistics in Human Genetics*. Arnold, London, 1998.
- [22] H.-C. So, J. S. H. Kwan, S. S. Cherny, and P. C. Sham. Risk prediction of complex diseases from family history and known susceptibility loci, with applications for cancer screening. *The American Journal of Human Genetics*, 88(5):548–565, 2011.
- [23] S. Wright. An analysis of variability in number of digits in an inbred strain of Guinea pigs. *Genetics*, 19(6):506–536, 1934.
- [24] Y. Wu et al. Pervasive biases in proxy genome-wide association studies based on parental history of Alzheimer’s disease. *Nature Genetics*, 56:2696–2703, 2024. doi:10.1038/s41588-024-01963-9.
- [25] Z. Zhuang et al. Relatedness, case-control imbalance and tail behaviour in family-history phenotypes. *Bioinformatics*, 2022. doi:10.1093/bioinformatics/btac459.